No.	Status	Date
QSCP-IP-ATU100 Rev 0	Issued	1995-12-04

Piping Inspection Plan - ATU-100

Scope and Objectives

This document establishes the initial inspection strategy for the piping systems within the ATU-100. The plan defines the requirements for monitoring internal and external degradation to ensure mechanical integrity throughout the service life of the unit. Douglas Ferreira is responsible for the technical oversight and implementation of these requirements in accordance with the QSCP Hamilton Inspection Department quality manual.

The scope encompasses all process piping identified in the unit circuitization, focusing on amine, reflux, and treated gas services. The primary objective is to establish a baseline for corrosion monitoring and to define the frequency of subsequent examinations based on the risk and service class of each circuit.

Regulatory and Technical Standards

The methodologies for inspection, repair, and alteration of the piping systems shall comply with the requirements set forth in API 570. This plan utilizes the guidelines for inspection practices and piping components as described in API 574 to ensure comprehensive coverage of all fittings and valves within the ATU-100.

All ultrasonic thickness measurements and non-destructive examinations must be performed following the specific protocols outlined in QSCP-NDT-001 Rev 0. This site-specific procedure ensures that data acquisition is consistent across all 118 CMLs identified for this unit.

Corrosion Monitoring and Data Management

A total of 118 CMLs have been distributed across the piping circuits to monitor for localized and general thinning. Douglas Ferreira shall ensure that each CML is permanently marked and recorded to allow for accurate long-term and short-term corrosion rate calculations. The retirement thickness for any component is defined as the line t-min.

In the event that a thickness reading is recorded below the established t-min, an NCR must be raised immediately. Corrosion rates will be determined by comparing the current data against the first reading or the latest pair of readings at least 3 years apart, whichever yields the larger value.

Inspection Intervals and Scheduling

Inspection intervals are determined by the remaining life calculation, where remaining life equals the current thickness minus t-min, divided by the corrosion rate. The next thickness reading must be scheduled at no more than half the remaining life, subject to the maximum intervals defined by the line class.

For Class 1 systems, including sour gas and regenerator overhead, the maximum interval is 5 years for both thickness and external visual inspections. Class 2 systems, such as amine and treated gas, are limited to 10 years for thickness and 5 years for external visual. Class 3 systems, including steam and cooling water, are limited to 10 years for both activities.

Responsibilities and Execution

The QSCP Hamilton Inspection Department is responsible for the execution of all NDE activities. Douglas Ferreira, as the API 570 inspector, is tasked with reviewing all field data and updating the inspection records. Any deviations from the requirements of QSCP-NDT-001 Rev 0 must be documented and approved prior to the completion of the inspection cycle.

Personnel performing the thickness measurements must be qualified to the standards required by the Hamilton Plant. The results of these inspections will form the basis for future adjustments to the ATU-100 inspection plan as the unit accumulates service history.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	1995-12-04	_____